# SNEHA SARAVANAN

**2+ Years as Software Engineer | AI Engineering, Deep Learning, Distributed Systems | Python, PyTorch, AWS, C++**
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## EDUCATION

**Carnegie Mellon University,** Pittsburgh, PA, USA December 2026
Master of Information Systems - Business Intelligence and Data *Analytics*

- Relevant Coursework: ML in Production (AI Engineering), AI Agents, Introduction to Deep Learning, Distributed Systems in Java
- **Teaching Assistantship: Introduction to Deep Learning**

**PES University,** Bengaluru, KA, India May 2024
Bachelor of Technology in Computer Science

- Relevant Coursework: Data Structures & Algorithms in C++, Machine Intelligence, Operating Systems, Computer Architecture
- **Publications: Springer Proceedings of International Conference on Computational Intelligence and Data Analytics** | URL

## SKILLS

**Languages:** Python, C++, SQL, Java, R
**ML/ DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Neural Networks, Drift Detection, A/B Testing, LangChain, Audio ML, Speech processing
**MLOPs/ Systems:** AWS (Bedrock, SageMaker, S3, Redshift), Docker, Kubernetes, Airflow, Kafka, Spark, GitHub Actions, Prometheus, Hadoop, MLFLow, Grafana
**Data & Visualization:** Pandas, NumPy, Plotly, Matplotlib, Seaborn, time series forecasting, Tableu, Power BI

## WORK EXPERIENCE

**Software Engineer - Data & ML| Eltropy, Bengaluru, India** July 2023 - August 2025

- Used Python, Scikit-learn, and AWS SageMaker to build a hybrid SARIMA + Prophet forecasting system with AWS CloudWatch monitoring for workforce planning, replacing manual staffing allocation for 100+ credit union clients
- Deployed a production LSTM-based loan forecasting model using TensorFlow and AWS SageMaker for end-to-end training and deployment, achieving 89% validation accuracy in production forecasting
- Utilized AWS Glue, Python, SQL, and Scikit-learn-based validation logic to build a data quality and drift detection framework with KPI checks and A/B testing, reducing data quality failures by 40%
- Engineered distributed ETL pipelines using Spark and Hadoop with SQL-based transformations for large-scale data processing, achieving zero downtime and eliminating schema drift in production systems
- Developed a production inference and deployment system using AWS SageMaker, Airflow, and Python for orchestrated ML workflows, enabling scalable model serving across 100+ enterprise clients

**Research Assistant | Carnegie Mellon University, Pittsburgh, PA, USA** October 2025 - Present

- Leveraged Python, SQL, Pandas, and NumPy to build a 200+ GB ML-ready data pipeline with validation and drift detection, reducing data provisioning time from 2 days to 4 hours
- Constructed data preprocessing pipelines for feature engineering and validation using Scikit-learn and Python reducing ML experimentation time by ~60% and ensuring reproducible ML workflows

## PROJECTS

**RedisLite – In Memory Key-Value Store | C++ / Systems Programming**

- Achieved a high-performance in-memory key-value store using C++ and epoll-based concurrency, achieving 100K+ ops/sec throughput and <1ms latency
- Implemented C++ custom arena allocator, LRU caching, and mutex-based concurrency control to optimize memory usage, reducing overhead by 35%
- Developed persistence and TTL eviction systems using C++ data structures (append-only logs, heap-based expiration) ensuring zero data loss and deterministic expiration under load

**Movie Recommendation System | ML in Production**

- Led the development of an LLM-based recommendation system using PyTorch, Kubernetes, and Kafka streaming pipelines with automated retraining workflows on event-driven data stored on SQLite
- Used Prometheus, Grafana, and Python monitoring scripts to build an observability stack for tracking drift, latency, and system health, achieving 99.2% system uptime
- Tested with Python and Scikit-learn statistical testing (A/B testing, t-tests) to evaluate ranking models across 1M users, achieving 23% cost reduction in LLM usage

**Audio Event Modelling & Speech Recognition System | Deep Learning, Transformers, PyTorch**

- Developed a Transformer-based audio event classification system using PyTorch and attention-based architectures, improving sequence prediction accuracy by 18%
- Built PyTorch beam search decoding and optimized sequence loss functions to reduce CER by 12–15% in speech recognition tasks
- Designed and engineered a real-time inference pipeline using Python and NumPy preprocessing systems for audio stream processing with <100ms latency